

# Thermal vs. Two-Break Spectra in Single-Pulse GRBs

What we are doing, why, and how — a primer

Prepared for Khushboo's presentation

Single-pulse GRB spectral programme

June 3, 2026

# What you are presenting — the big picture

## One sentence

We take a clean sample of **single-pulse GRBs**, fit their **time-resolved** spectra with a family of physical models, and ask: *is the curvature in the spectrum a thermal (photosphere) component, or a second synchrotron break?*

The talk has six parts:

- ① **Why** thermal + non-thermal modelling (and why “two breaks”)
- ② **The sample** — Busby & Lazzati single-pulse selection
- ③ **How many** bursts — 106 single-pulse GRBs
- ④ **Background** — why we model it physically
- ⑤ **Bayesian blocks** — how we choose the time bins
- ⑥ **The cutoff** — Ravasio, time-resolved, and LAT/LLE

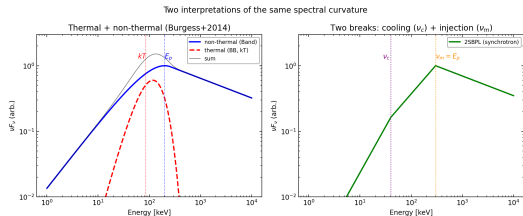
# 1. Why model a thermal *and* a non-thermal component?

The standard Band / cutoff-power-law (CPL) fit is **empirical** — it describes the curve but carries no physics.

Two physical ingredients are expected in the prompt emission:

- **Photosphere** → a quasi-thermal **blackbody (BB)**, temperature  $kT$ .
- **Optically-thin synchrotron** → a non-thermal continuum peaking at  $E_p$ .

**Burgess+2014** showed many bursts need *both*: a BB sitting under a non-thermal Band/synchrotron component. The two pieces move *together* in time — which is the key physical clue.



The same curvature, two readings: thermal + non-thermal (left) or two synchrotron breaks (right).

## 1. Burgess+2014: the thermal–non-thermal correlation

For each burst, Burgess fit **BB + non-thermal** in every time bin and plotted  $E_p$  against  $kT$ :

$$E_p \propto kT^\alpha$$

- $\alpha \approx 1 \rightarrow$  **baryonic** (matter-dominated) jet
- $\alpha \approx 2 \rightarrow$  **magnetic** (Poynting-dominated) jet

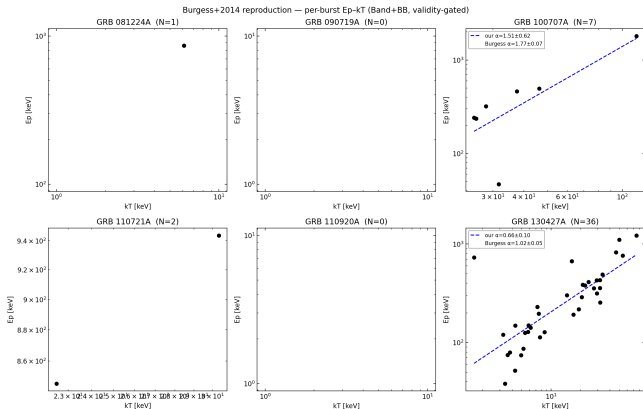
Combined sample: Spearman  $\rho = 0.81$ ,  
 $p = 4 \times 10^{-20}$  — a *very* tight link between the thermal and non-thermal peaks.

This correlation is the headline physics we want to reproduce and extend with a larger, uniformly-selected sample.

| GRB     | $\alpha$ (Burgess) | jet      |
|---------|--------------------|----------|
| 081224A | $1.01 \pm 0.14$    | baryonic |
| 090719A | $2.33 \pm 0.27$    | magnetic |
| 100707A | $1.77 \pm 0.07$    | magnetic |
| 110721A | $1.24 \pm 0.11$    | baryonic |
| 110920A | $1.97 \pm 0.11$    | magnetic |
| 130427A | $1.02 \pm 0.05$    | baryonic |

Burgess+2014, ApJL **784**, L43, Table 1.

# 1. Our reproduction of Burgess (so far)



All 6 of Burgess's bursts are in our sample. Using our pipeline's BB+non-thermal fits, with  $E_p$  taken as the *true*  $\nu F_\nu$  peak of each model:

- **130427A:**  $\alpha = 0.66$ , baryonic — *matches*
- **100707A:**  $\alpha = 1.51$ , magnetic — *matches*
- Combined:  $\rho = 0.69$ ,  $p = 1.5 \times 10^{-7}$  (his 0.81)

The other 4 have only  $N \leq 2$  time bins with a *significant* BB — a statistics limit, not a contradiction. A larger sample fixes this.

## 2. The other reading: two synchrotron breaks

A pure-synchrotron spectrum from a cooling electron population has **two** characteristic break energies:

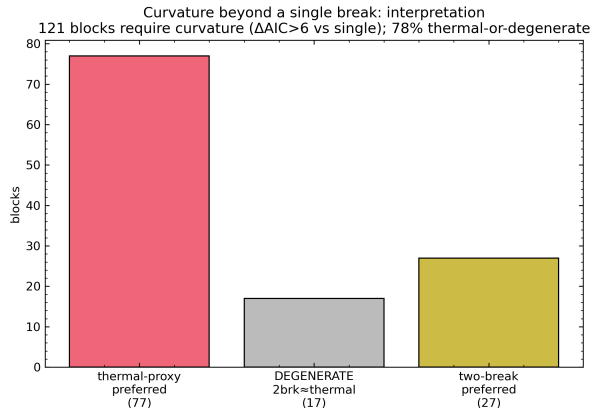
- $\nu_c$  — the **cooling break** (electrons that have radiated away their energy).
- $\nu_m = E_p$  — the **injection / peak** energy (minimum electron Lorentz factor).

A **double smoothly-broken power law (2SBPL / DSBPL)** captures both breaks. This is the model Ravasio and others use to argue prompt emission *is* synchrotron.

### The key idea in this project (Vikas)

A **BB + SBPL** (thermal + one-break) fit is an effective *proxy* for a **2SBPL** (two-break): both add curvature near the peak. So we fit *both* families and let the data decide which is genuinely required.

## 2. What the sample says: proxy vs. genuine second break



Of **121** time-bins that need curvature beyond a single break ( $\Delta AIC > 6$ ):

- **77** prefer the thermal proxy (BB+SBPL)
- **17** are degenerate (2-break  $\approx$  thermal)
- **27** genuinely prefer the explicit two-break

⇒ **~78% thermal-or-degenerate**: BB+SBPL really is a good proxy for 2SBPL, exactly as expected. Only **~22%** *demand* the second break.

Standout genuine two-break burst: **130427A** (brightest GRB ever) — one bin prefers DSBPL by  $\Delta AIC = 590$ .

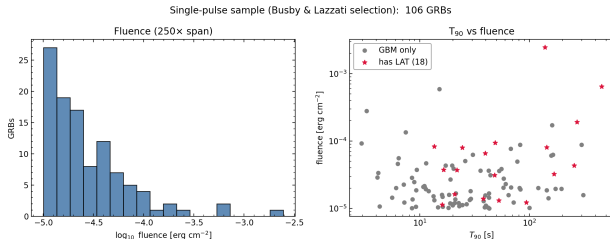
### 3. The sample: Busby & Lazzati single-pulse selection

**Why single-pulse?** A single, clean pulse is the simplest physical unit — no pulse overlap to confuse the spectral evolution. Ideal for testing thermal vs. synchrotron.

**Selection (Busby & Lazzati 2024, ApJ 972, 83):**

- Fit each GBM burst's light curve for pulse shape.
- Score how well a *single* pulse describes it.
- Keep bursts above a strict threshold ( $\gtrsim 0.9983$ ).

**Important:** this selects on *shape*, *not* brightness — so fluence spans a factor of  $\sim 250$  across the sample.





### 3. How many bursts — 106 single-pulse GRBs

- **106** single-pulse GRBs fully analysed <sup>a</sup>
- **1169** time-resolved spectral bins in total.
- **18** have high-energy **LAT** coverage (red stars).
- Every burst fit with **all available detectors**:  
Nal ( $\leq 50^\circ$ ), BGO, and LAT-LLE where present.

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<sup>a</sup>Vikas's note says "105"; our current pipeline has 106 — one near-threshold burst. Worth confirming the exact count.

#### Six models per time-bin:

- 1 Band
- 2 Cutoff power law (CPL)
- 3 Smoothly-broken PL (SBPL)
- 4 Double SBPL (DSBPL = two breaks)
- 5 Band + Blackbody
- 6 CPL + Blackbody

All fit inside **3ML** (the Multi-Mission Max-Likelihood framework), with `pgstat` likelihood (Poisson source, Gaussian background). Model choice by AIC/BIC + a likelihood-ratio test, with a physical-validity gate.

## 4. Why we model the background physically

GBM has **no off-source pointing** — the detectors always see the whole sky, so the background is large and *time-varying* (it drifts with the spacecraft's orbit and geomagnetic position).

### What we do, per detector:

- 1 Pick **pre-burst** and **post-burst** windows, each  $\sim 50\text{--}150$  s wide (wide enough to constrain, narrow enough to avoid orbital trends).
- 2 Fit a low-order **polynomial** in time to the count rate in those windows (this is the same algorithm `gtburst` / 3ML uses).
- 3 Interpolate that polynomial *under* the burst to get the background at each source time.

### Why it matters

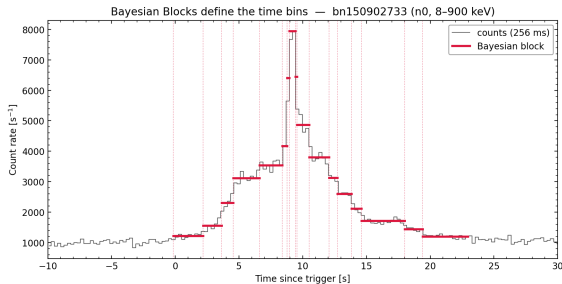
A wrong background biases every spectral parameter — especially the low-energy slope and any faint thermal component. Getting the background right is *the* prerequisite for trusting a BB detection. Each detector gets its own background; selections are reviewable by a human.

## 5. How the time bins are made: Bayesian Blocks

We do **not** use fixed-width time bins. Instead, **Bayesian Blocks** (Scargle's algorithm, `astropy.stats.bayesian_blocks`) find the optimal change-points where the count rate *statistically* changes.

- Run on the background-subtracted events, **8–900 keV**.
- `fitness='events'` on the photon arrival times; a false-alarm prior  $p_0$  sets how eager it is to split.
- Each block becomes one **spectral fit** — so bins are fine where the spectrum is changing fast, coarse where it is steady.

For a handful of ultra-bright bursts ( $>10^6$  events) we use the *binned* variant (`fitness='measures'`) to avoid an  $O(N^2)$  blow-up.



Red = Bayesian blocks; dotted lines = the change-points that become our time-bin edges.

## 6. The cutoff / extra break — Ravasio, and our extension

**Ravasio+2018/2019** found that many GRB spectra show a *low-energy break* (slopes  $-2/3$  then  $-3/2$ ) on top of the peak — the smoking gun of **synchrotron with a cooling break**. Crucially, their work was largely **time-integrated** (or coarse bins).

### Our extension:

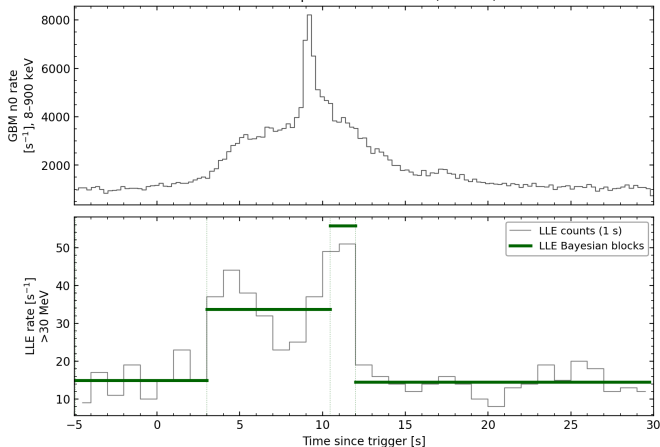
- Do the same break/cutoff search **time-resolved**, bin by bin (this is what the Bayesian blocks give us).
- Use **LAT / LAT-LLE** ( $>30$  MeV) to anchor the *high-energy* end and constrain any high-energy cutoff — the lever arm GBM alone lacks.

### Why LAT/LLE matters for the cutoff

A high-energy cutoff or the upper synchrotron break is only constrained if you have photons up there. The LAT/LLE light curve tells us *when* high-energy emission is present — i.e. which time bins can actually constrain the cutoff.

## 6. LAT/LLE Bayesian-blocks light curve

bn150902733: keV pulse vs.  $>30$  MeV (LAT-LLE) emission



**bn150902733** (a LAT burst):

- Top: the keV pulse (GBM).
- Bottom: the  $>30$  MeV (LLE) light curve, with its *own* Bayesian blocks.

The high-energy emission is concentrated in  $\sim 3$ – $12$  s — so *those* time bins are where we can put real constraints on a high-energy cutoff or upper synchrotron break. Outside that window there are too few MeV photons.

This is the template for the time-resolved cutoff study across all 18 LAT bursts.

- ➊ **Physics question:** is GRB spectral curvature a *thermal* photosphere (BB) or a *second synchrotron break* (2SBPL,  $\nu_c$  &  $\nu_m$ )? Both are physical; we let the data choose.
- ➋ **Sample:** 106 *single-pulse* GRBs (Busby & Lazzati, shape-selected), 1169 time bins, 18 with LAT.
- ➌ **Method:** background modelled physically per detector; Bayesian blocks set the time bins; 6 models fit in 3ML; AIC/BIC + LRT decide.
- ➍ **Result so far:**  $\sim 78\%$  of curved bins are thermal-or-degenerate  $\Rightarrow$  BB+SBPL is a good proxy for 2SBPL;  $\sim 22\%$  genuinely need two breaks. We recover Burgess's  $kT-E_p$  correlation ( $\rho = 0.69$ ).
- ➎ **Next:** time-resolved cutoff search (extending Ravasio), using LAT/LLE to anchor the high-energy end.

- Burgess, J.M. et al. 2014, *ApJL* **784**, L43 — thermal + non-thermal,  $kT-E_p$  correlation.
- Busby, K. & Lazzati, D. 2024, *ApJ* **972**, 83 — single-pulse GRB sample selection.
- Ravasio, M.E. et al. 2018 (*A&A* 613, A16) & 2019 (*A&A* 625, A60) — synchrotron low-energy break / cutoff.
- Scargle, J.D. et al. 2013, *ApJ* 764, 167 — Bayesian blocks.
- Vianello, G. et al. 2015 — 3ML (the fitting framework).

Good luck, Khushboo!